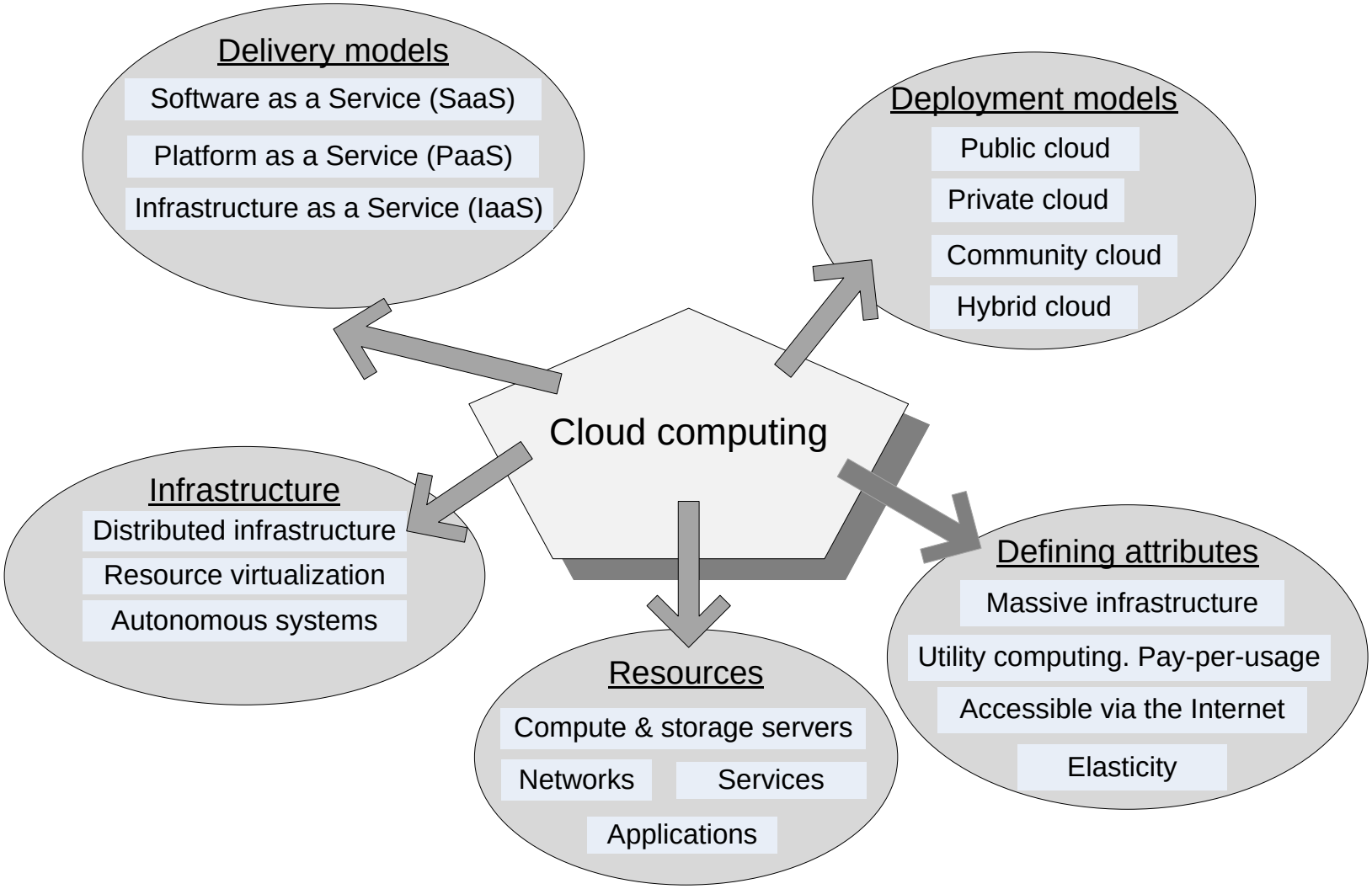
What is Cloud Computing?

- "**Cloud computing** is an information technology (IT) paradigm that enables ubiquitous access to shared pools of configurable system resources and higher-level services that can be rapidly provisioned with minimal management effort, often over the Internet. Cloud computing relies on sharing of resources to achieve coherence and economies of scale, similar to a public utility." https://en.wikipedia.org/wiki/Cloud_computing

- "Simply put, cloud computing is the delivery of computing services – servers, storage, databases, networking, software, analytics and more – over the Internet ("the cloud"). Companies offering these computing services are called cloud providers and typically charge for cloud computing services based on usage, similar to how you're billed for gas or electricity at home." https://azure.microsoft.com/en-gb/overview/what-is-cloud-computing/

Cloud Computing Models, Resources, Attributes



Early Models of Cloud Computing

- Basic reasoning: information and data processing can be done more efficiently on large farms of computing and storage systems accessible via the Internet.
- Two early models:

1. **Grid computing** – initiated by the National Labs in the early 1990s; targeted primarily at scientific computing.
   - *"Grid computing is the collection of computer resources from multiple locations to reach a common goal. The grid can be thought of as a distributed system with non-interactive workloads that involve a large number of files."* from Wikipedia
2. **Utility computing** – initiated in 2005-2006 by IT companies and targeted at enterprise computing.
   - *"Utility computing is a service provisioning model in which a service provider makes computing resources and infrastructure management available to the customer as needed, and charges them for specific usage rather than a flat rate."* from Wikipedia

Cloud computing - Characteristics

"*Cloud Computing offers on-demand, scalable and elastic computing (and storage services). The resources used for these services can be metered and users are charged only for the resources used.* " from the Book

Shared Resources and Resource Management:

1. Cloud uses a shared pool of resources
2. Uses Internet techn. to offer **scalable** and **elastic** services.
3. The term "**elastic computing**" refers to the ability of **dynamically** and **on-demand** acquiring computing resources and supporting a variable workload.
4. Resources are metered and users are charged accordingly.
5. It is more cost-effective due to **resource-multiplexing.** Lower costs for the cloud service provider are past to the cloud users.

Cloud computing (cont' d)

Data Storage:

6. Data is stored:
   - in the "cloud", in certain cases closer to the site where it is used.
   - appears to the users as if stored in a location-independent manner.
7. The data storage strategy can increase reliability, as well as security, and can lower communication costs.

Management:

8. The maintenance and security are operated by service providers.
9. The service providers can operate more efficiently due to specialisation and centralisation.

Cloud Computing Advantages

1. Resources, such as CPU cycles, storage, network bandwidth, are **shared.**

2. When multiple applications share a system, their peak demands for resources are not synchronised thus, **multiplexing** leads to a higher resource utilization.

3. Resources can be **aggregated** to support data-intensive applications.

4. Data sharing facilitates **collaborative** activities. Many applications require multiple types of analysis of shared data sets and multiple decisions carried out by groups scattered around the globe.

Cloud Computing Advantages

5. Eliminates the **initial investment costs** for a private computing infrastructure and the maintenance and operation costs.

6. **Cost reduction**: concentration of resources creates the opportunity to pay as you go for computing.

7. **Elasticity**: the ability to accommodate workloads with very large peak-to-average ratios.

8. **User convenience**: virtualization allows users to operate in familiar environments rather than in idiosyncratic ones.

Types of clouds

1. **Public Cloud** - the infrastructure is made available to the general public or a large industry group and is owned by the organization selling cloud services.

2. **Private Cloud** – the infrastructure is operated solely for an organization.

1. **Hybrid Cloud** - composition of two or more Clouds (public, private, or community) as unique entities but bound by a standardised technology that enables data and application portability.

2. **Other types: e.g., Community/Federated Cloud** - the infrastructure is shared by several organizations and supports a community that has shared concerns.

Challenges for cloud computing

1. Availability of service: what happens when the service provider cannot deliver?

2. Data confidentiality and auditability, a serious problem.

3. Diversity of services, data organization, user interfaces available at different service providers limit user mobility; once a customer is hooked to one provider it is hard to move to another.

4. Data transfer bottleneck; many applications are data-intensive.

More challenges

5. Performance unpredictability, one of the consequences of resource sharing.
   - How to use resource virtualization and performance isolation for QoS guarantees?
   - How to support elasticity, the ability to scale up and down quickly?

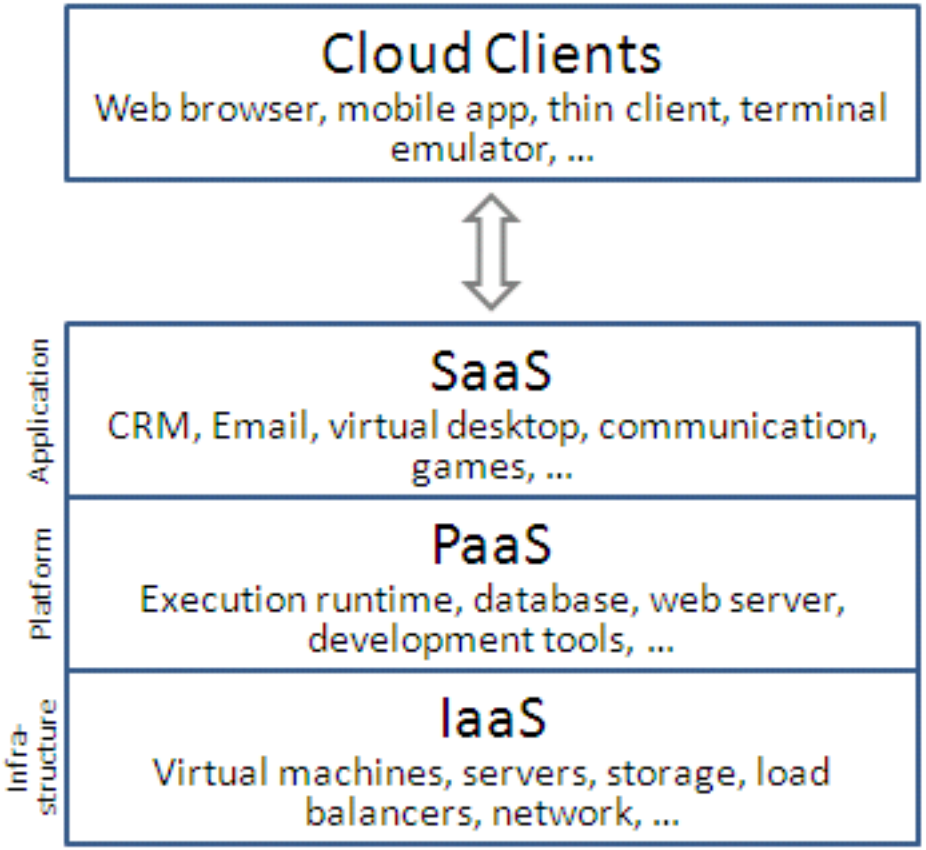
6. Resource management: It is a big challenge to manage different workloads running on large data centers. Are self-organization and self-management the solution?

7. Security and confidentiality: major concern for sensitive applications, e.g., healthcare applications.

Addressing these challenges is on-going work!

Cloud Delivery Models

1. **Software as a Service (SaaS)** (high level)
2. **Platform as a Service (PaaS)**
3. **Infrastructure as a Service (IaaS)** (low level)



source Wikipedia

Infrastructure-as-a-Service (IaaS)

- Infrastructure is compute resources, CPU, VMs, storage, etc
- The user is able to deploy and run arbitrary software, which can include operating systems and applications.
- The user does not manage or control the underlying Cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of some networking components, e.g., host firewalls.
- Services offered by this delivery model include: server hosting, storage, computing hardware, operating systems, virtual instances, load balancing, Internet access, and bandwidth provisioning.
- Example: Amazon EC2

Platform-as-a-Service (PaaS)

- Allows a cloud user to deploy consumer-created or acquired applications using programming languages and tools supported by the service provider.
- The user:
  - Has control over the deployed applications and, possibly, application hosting environment configurations.
  - Does not manage or control the underlying Cloud infrastructure including network, servers, operating systems, or storage.
- Not particularly useful when:
  - The application must be portable.
  - Proprietary programming languages are used.
  - The hardware and software must be customised to improve the performance of the application.
- Examples: Google App Engine, Windows Azure

Software-as-a-Service (SaaS)

- Applications are supplied by the service provider.
- The user does not manage or control the underlying Cloud infrastructure or individual application capabilities.
- Services offered include:
  - Enterprise services such as: workflow management, communications, digital signature, customer relationship management (CRM), desktop software, financial management, geo-spatial, and search.
- Not suitable for real-time applications or for those where data is not allowed to be hosted externally.

- Examples: Gmail, Salesforce

The Three delivery models of Cloud Computing

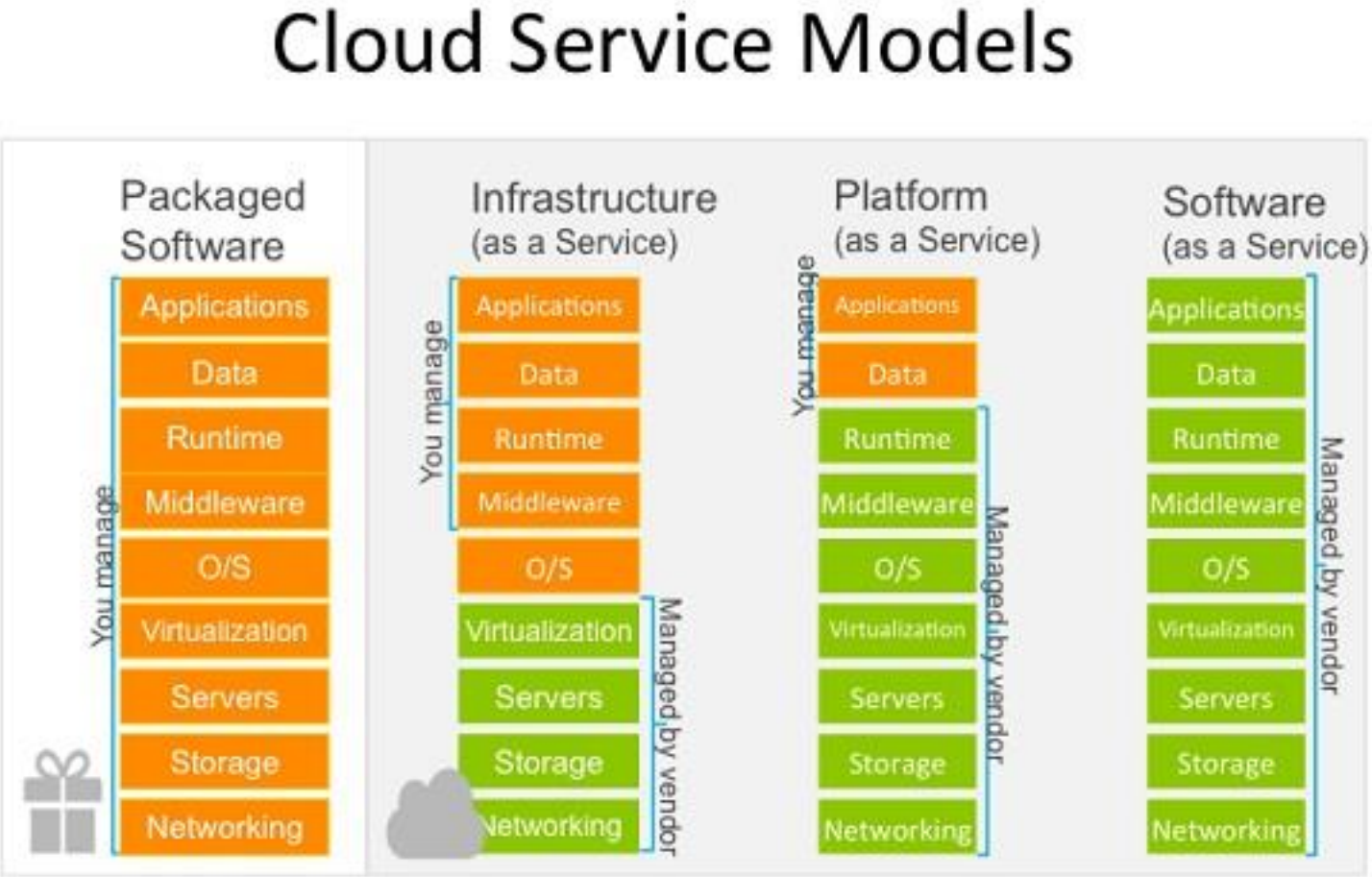


Figure 1. **Source: Microsoft Azure**

Cloud activities

- Service management and provisioning including:
  - Virtualization.
  - Service provisioning.
  - Call center.
  - Operations management.
  - Systems management.
  - QoS management.
  - Billing and accounting, asset management.
  - SLA management.
  - Technical support and backups.